

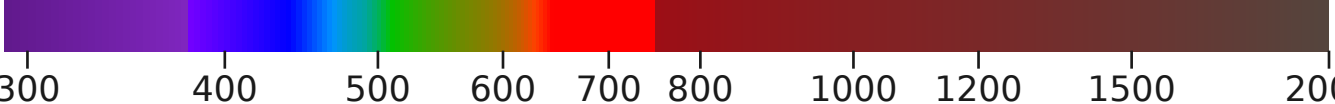
Polonium

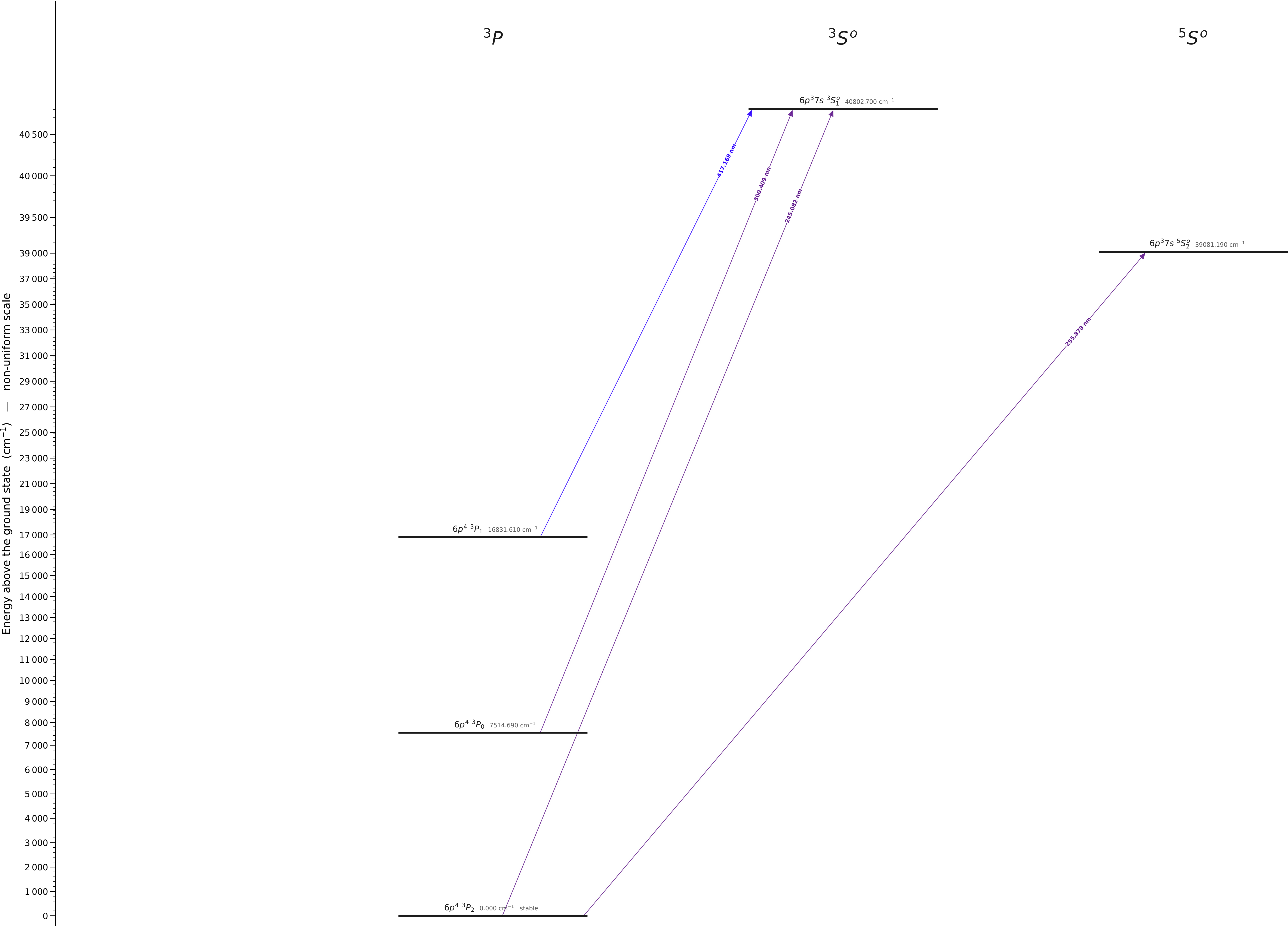
Po (Po I). Strongest classified transitions below 2 μm (NIST): 5 levels, 4 lines.

Po⁺ ionisation limit 67896.31 cm⁻¹ = 8.41807 eV (147.283 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J \| e r \| J' \rangle |$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)





Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
$6p^4 \ ^3P_2 - 6p^3 7s \ ^5S_2^o$	255.8776	255.8008	1171.6246	-	-	-	-	first step of the ISOLDE-RILIS polonium resonance ionizat...

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

²⁰⁹Po hyperfine structure (*I* = 1/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$6p^3 7s \ ^5S_2^o$	2127(3)	-	F=3/2: -3190.50 F=5/2: +2127.00

Measured A, B constants; sources in the data file.